The Estimator Is the Curriculum:

Frontier Sampling and Failure Recycling for Likelihood-Based RL

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Code, logs, and proofs: https://github.com/linjiw/curriculum-maxrl

Abstract

Reinforcement learning with verifiable rewards spends most of its budget generating rollouts, and on hard task pools most of that budget buys nothing: under uniform prompt sampling, we measure that **65–75% of rollout groups produce zero learning signal** — every rollout fails, and success-conditioned estimators drop the group. The emerging fix is a difficulty curriculum, but current curricula are heuristics bolted on from outside: bandit scores, hand-set difficulty bands, filtering rules. We show the curriculum is already inside the estimator. For MaxRL-style success-conditioned advantages, the expected learning signal a prompt emits from a group of N rollouts has a closed form, $\mathbb{E}[\sum_i |w_i|] = 2\,(\text{pass@}N - \text{pass@}1)$ — a compute-indexed zone-of-proximal-development functional, maximal on prompts solvable within N attempts but not within one, peaking at pass rate $\approx \ln N/N$. Sampling prompts by this quantity (a decayed Thompson posterior makes it practical) turns the rollout budget N you already chose into the curriculum knob, with no new difficulty hyperparameters; published learnability curricula fall out as its $N{=}2$ slice. The same algebra draws a hard boundary: no sampling rule can rescue an all-fail prompt, because allocation only redistributes signal that exists. We therefore add a second, complementary mechanism — **failure recycling**: a failed trajectory is a verified success for the goal it actually reached, and relabeling dead groups (with the prompt's goal conditioning rewritten) provably yields the maximum-likelihood gradient of the relabeled task, exact when conditional laws match. On testbeds spanning exact-gradient chains, a 1.26M-parameter maze transformer, Gymnasium control, and a 360M LLM on GSM8K, the combined schedule beats a true-pass-rate **oracle** allocator (0.890 vs. 0.851 AUC), turns a frontier-heavy regime from unlearnable to solved ($0 \to 0.98$ at equal compute), grows pass@k coverage where GRPO's collapses, and needs up to 11× fewer inference samples to reach target coverage. A pre-registered LLM-scale 2×2 confirms the safety half of the claim: the identical curriculum that trains stably under MaxRL *degrades* GRPO — curricula amplify objective-level pathologies, so the estimator underneath decides whether a curriculum is safe at all.

1 Introduction

Post-training language models with verifiable rewards has a cost structure unlike ordinary supervised learning: the data is free, but *rollouts* are not. Every optimization step is preceded by sampling N completions per prompt, and generation dominates wall-clock (86% of step time in our LLM runs). What that compute buys depends entirely on where it lands. On a prompt the model has mastered, N successes carry no contrast and no gradient. On a prompt beyond the model's reach, N failures carry no successes — and for the success-conditioned estimators of likelihood-based RL, the group is dropped entirely. We measure the combined waste at 65–75% of groups under uniform sampling. The field's response is difficulty curricula — but the current generation derives its sampling rule from outside the learner: UCB bandits over difficulty buckets [2], target-difficulty controllers [3], category bandits [4], rejection rules [5], or dead-prompt filtering [6, 7]. Each adds machinery and hyperparameters to estimate, in effect, *where learning is possible right now.*

Our starting observation is that likelihood-based RL already computes this. MaxRL [1] showed that standard RL on binary rewards optimizes a first-order approximation of maximum likelihood, and that normalizing advantages by the group's success count K instead of its size N recovers a truncated-ML objective. Reading their weight-function view raises a question with a non-obvious answer: *if the objective already pours gradient into hard prompts, is there anything left for a curriculum to do?* The answer is sharply yes, for a structural reason: **weights act on prompts after they are sampled, and only when at least one**

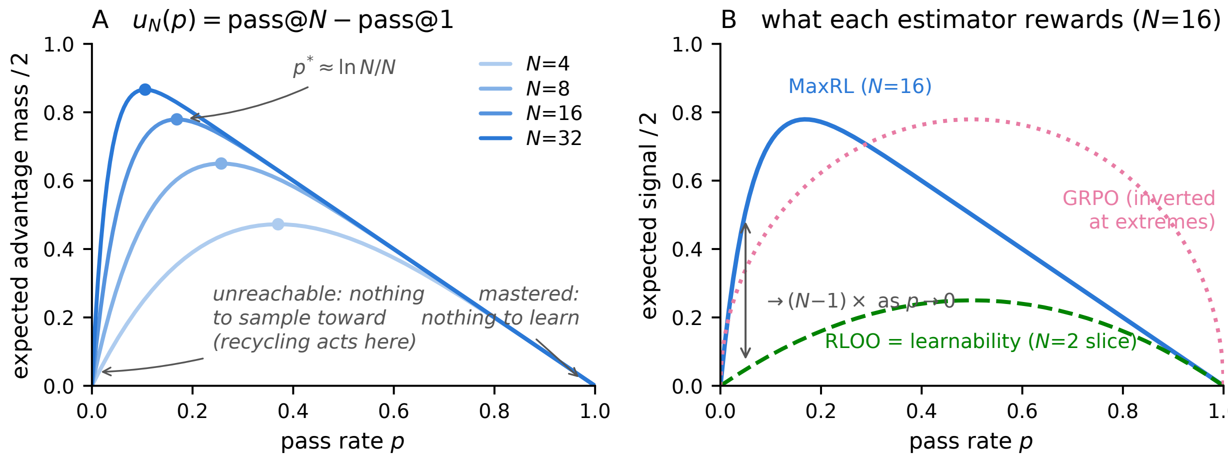

Figure 1: **The estimator is the curriculum.** *Left:* the derived utility $u_N(p) = (1-(1-p)^N) - p$ — the estimator's expected advantage mass (Prop. 1) — for growing rollout budgets N. The peak $p^* \approx \ln N/N$ walks toward harder prompts as compute grows; both tails are dead zones the weight function cannot reach. *Right:* at N=16, MaxRL concentrates $\to (N-1)\times$ more expected signal than RLOO on frontier prompts (Prop. 2); RLOO's profile is exactly the published "learnability" objective — the N=2 slice.

rollout succeeds. However the weight function is shaped, it cannot rescue a group that came back all-fail — dropping that group is precisely what makes the estimator unbiased — and it cannot un-spend the compute burned re-confirming mastered prompts. The estimator's blind spots define the curriculum's job description.

Taking that sentence literally produces this paper. Our contributions:

1. **The curriculum is a theorem, not a heuristic** (§3): the expected advantage mass of a prompt is exactly $2(\text{pass@}N - \text{pass@}1)$, a derived ZPD functional whose band location, width, and compute-scaling are set by N alone. Learnability curricula are its N=2 slice; optimal rollout allocation is water-filling on $p(1-p)^N$.
2. **A practical teacher** (§4): Thompson sampling on a decayed Beta posterior with the derived utility; the estimator is untouched, so its unbiasedness carries over.
3. **Failure recycling with an exactness guarantee** (§5): relabeling dead groups to achieved sub-goals yields the ML gradient of the relabeled task; exact when conditional laws match, measured indistinguishable from fresh groups (cosine 0.956 vs. 0.958), with two contracts whose violation we price.
4. **A three-channel account with a safety rule** (§6–7): allocation is bounded by an oracle ceiling; creation breaks it; the objective underneath decides whether either is safe — confirmed at LLM scale on a pre-registered prediction.
5. **An honest evidence discipline**: pre-registered predictions, documented negatives with diagnoses, one retraction, and a two-round adversarial audit tracing every number to a log or proof.

2 Background: MaxRL in three lines

For binary rewards, maximum likelihood maximizes $\mathbb{E}[\log p_\theta(\text{success} \mid x)]$. MaxRL expands $\log p$ as a Maclaurin series in $(1-p)$ and truncates at order T, giving a compute-indexed family $J^{(T)}$ interpolating REINFORCE (T=1) to exact ML ($T \to \infty$); the practical estimator sets per-rollout advantages $w_i = r_i/K - 1/N$ with all-fail groups dropped, which is unbiased for the $T = N-1$ objective. Its weight function grows as $p \to 0$: the objective is an *implicit, gradient-level* curriculum. Our work is the data-level complement: the same algebra, read as a sampling rule plus a recycling rule.

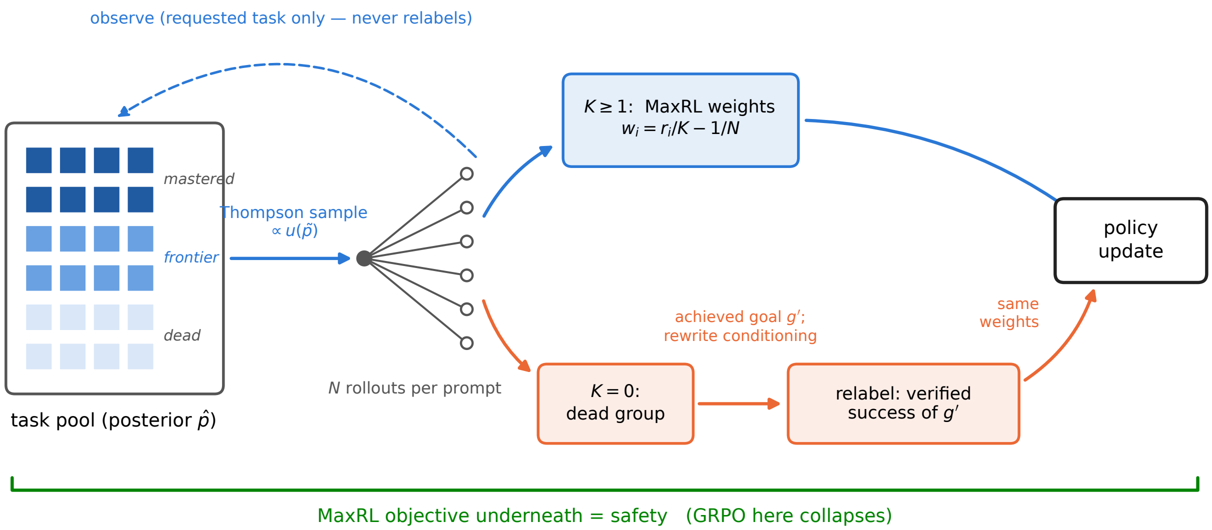

Figure 2: **One FrontierMax step.** A Thompson-sampled posterior allocates rollouts by the derived utility; live groups train with unmodified MaxRL weights; dead groups are relabeled to the sub-goal they actually achieved (conditioning rewritten) and train as verified successes. The posterior observes requested-task outcomes only.

3 The estimator is the curriculum

Proposition 1 (Advantage mass; MC-verified)**.** *For a prompt with pass rate p and N i.i.d. rollouts under the practical MaxRL weights,*

$$\mathbb{E}\Big[\sum_i |w_i|\Big] \;=\; 2\big(pass@N(p) - pass@1(p)\big) \;=\; 2\big((1-(1-p)^N) - p\big).$$

Interpretation. *The estimator's expected learning signal on a prompt is twice the probability that the prompt is solvable within N attempts but not within one. That sentence is a curriculum: zero on mastered prompts, zero on unreachable ones, maximal at $p^* = 1 - N^{-1/(N-1)} \approx \ln N/N$ — a zone of proximal development whose center, width, and compute-scaling are set by the one parameter you already chose.*

Proposition 2 (Frontier amplification)**.** *As $p \to 0$, the ratio of MaxRL's expected signal to RLOO's ($2p(1-p)$ exactly) tends to $N-1$.*

Interpretation. *Concentrating sampling on the frontier only helps if the estimator can extract signal there; GRPO's variance-normalized weights invert the profile, which §7 shows becoming an active failure under a curriculum.*

Proposition 3 (Allocation)**.** *The rollout allocation maximizing total expected signal is greedy water-filling on the marginal $p(1-p)^N$ — the probability that the next rollout is a group's first success.*

One identity, three literatures: learnability curricula [5, 8] are the N=2 slice of u_N; DAPO-style dynamic sampling and GRESO are its avoidance shadow (cull where $u \approx 0$); HER-style relabeling is its creation complement (manufacture $K > 0$ where $u = 0$ identically; §5).

4 FrontierMax: the schedule

A decayed Beta posterior (decay 0.7) tracks each prompt's pass rate from observed group outcomes only — never from relabeled successes (violating this inflates the posterior: $\hat{p}$ 0.81 vs. eval 0.47). Thompson sampling draws $\tilde{p}$ and samples prompts $\propto u(\tilde{p})^\gamma$ with a 10% uniform floor; $\gamma \approx 4$ when tasks share skills

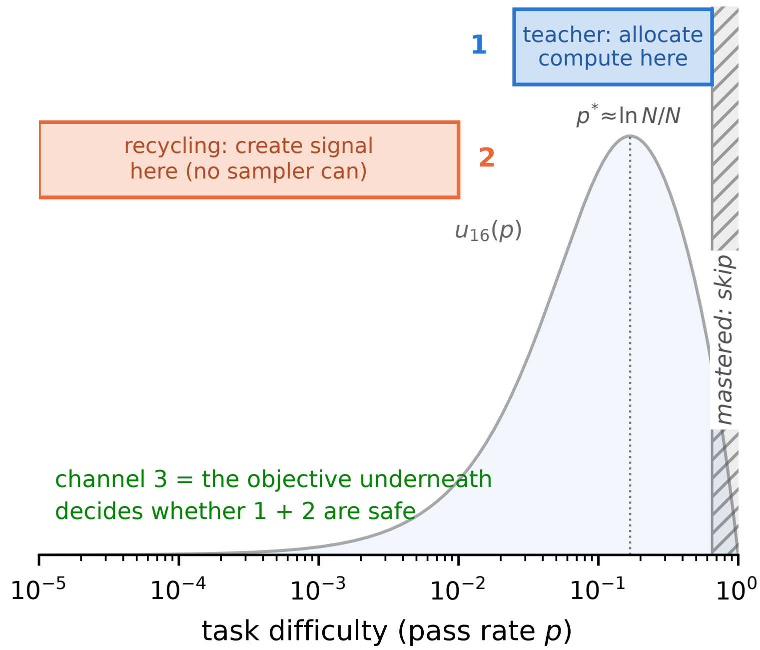

Figure 3: **Three channels.** The teacher allocates compute to the frontier band; recycling creates signal in the dead zone no sampler can reach; the objective underneath decides whether either is safe.

(learning compounds — validated on chains, and its non-transfer to broad pools was predicted in advance by a compounding ODE model), $\gamma = 1$ otherwise. Live groups train with unmodified MaxRL advantages. Honest knob inventory: decay, floor, and γ exist, with validated defaults; what is *derived* rather than tuned is the difficulty band itself — location, width, and N-scaling — which is exactly where competing curricula spend their hyperparameters.

5 Failure recycling

Proposition 4 (Hindsight exactness)**.** *Relabeling a dead group to the sub-goal g' its trajectories actually achieved, rewriting the goal conditioning, and applying the same success-conditioned weights yields the ML gradient of task g' under the conditional law shifted by the original sampling; the two coincide exactly when the conditional laws match.*

Interpretation. *Measured: per-group cosine to the true relabeled-task gradient 0.956, vs. 0.958 for fresh unbiased groups; the mean relabeled gradient reaches cosine 1.000.*

Two contracts keep practice on the proof's side: **(1) exactness** — a relabeled success must be a true success of the relabeled task under the environment's own verifier (never an LLM judge); **(2) conditioning** — goal-embedding trajectories must have their conditioning rewritten to the achieved goal. Skipping (2) turns the exact gradient into noise: on our gridworld, hindsight without the rewrite scores below teacher-only ($0.600 < 0.658$); with it, it leads (0.703). Recycling is the only mechanism here that *creates* signal, and its value is regime-dependent in a way we can state precisely: the gain is proportional to how much a relabeled skill can compound — +0.22 AUC on fixed task pools, +0.01 on one-shot task streams. Task-set fixedness is the single most important regime variable we found.

6 Three channels, one safety rule

The **teacher** avoids waste — bounded by an oracle ceiling (a perfect allocator collects only 0.4% more advantage mass than our posterior). **Recycling** creates signal — the only channel that breaks that ceiling.

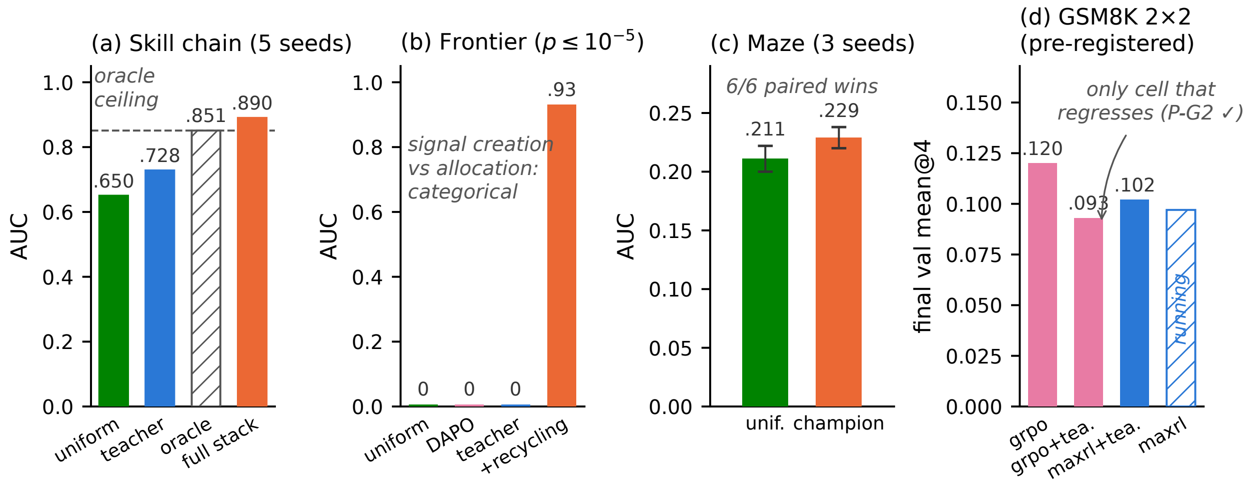

Figure 4: **The ladder.** (a) Exact-gradient chains: the full stack beats the true-pass-rate oracle. (b) Frontier-heavy regime: allocation scores exactly zero; creation solves it. (c) Maze at matched wall-clock: 6/6 paired seed wins. (d) GSM8K 2×2: the identical teacher helps MaxRL and hurts GRPO (pre-registered P-G2).

The **objective** underneath decides whether either is safe. One line: *the teacher allocates, hindsight creates, the objective decides whether either is safe.*

7 Experiments: an escalating ladder

7.1 Exact-gradient skill chains (CPU, 5 seeds). Uniform 0.650 → teacher 0.728 → full stack $\mathbf{0.890 \pm 0.002}$ — above the true-pass-rate oracle allocator (0.851) and above a sharper γ=4 oracle (0.884).

Takeaway. Allocation saturates; creation breaks the ceiling.

7.2 Frontier-heavy regime (max pool pass rate 10^{-5}). Uniform, DAPO, and the plain teacher score **exactly 0.00**; teacher+recycling reaches **0.98 final (0.93 AUC)** — relabeling invents the curriculum below the pool, ignites within ∼400 groups, then goes silent.

Takeaway. Where there is nothing to sample toward, only signal creation works — categorically, not marginally.

7.3 Maze transformer at matched wall-clock (1.26M params, 13 levels, ∼30 runs, 3 seeds). Champion (teacher + dense recycling) 0.252±0.005 final / 0.229±0.009 AUC vs. uniform 0.230±0.015 / 0.211±0.011; paired deltas positive 6/6 seeds; 22–35% more optimization steps per GPU-hour. Safety: the identical teacher grows pass@8 under MaxRL in every seed (0.316 → 0.348) while GRPO's coverage decays in every seed (0.308 → 0.271); the teacher-arm run amplifies the collapse (0.332 → 0.269; single-seed arm). Inference currency: 1.2 × /2.7 × /**11**× fewer samples than GRPO to reach target coverage at levels 2/3/5 (honest 0.5× reversal at level 4).

Takeaway. The gains survive real gradients; the safety warning is real; coverage is the currency that sees both.

7.4 Gymnasium control (MountainCar). Flag-only training: 0.000 — the classic exploration wall. The corrected 10-seed transition-matched study (paired bootstrap): flag pass uniform 0.058±0.079 → teacher 0.664±0.232 → full stack **0.848±0.058**; per-bin policy parameters stay at 0.000.

Takeaway. Curricula operate through shared parameters, or not at all.

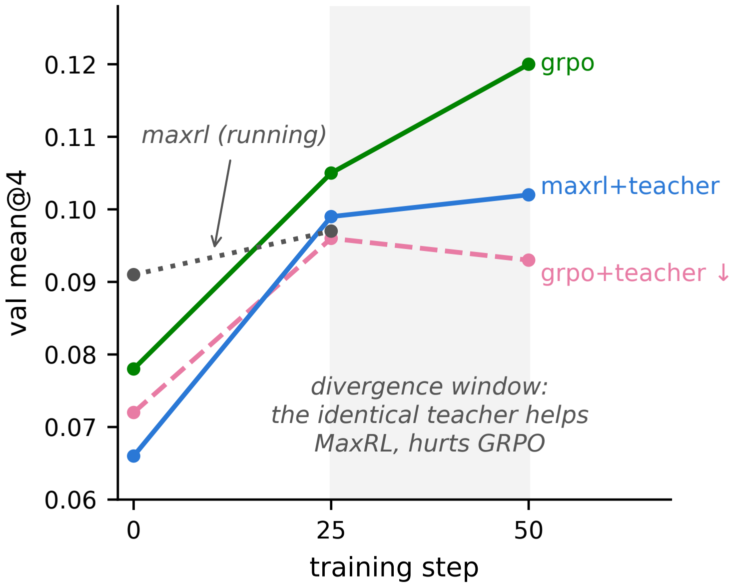

Figure 5: **GSM8K 2×2 (pre-registered).** The identical teacher helps MaxRL and hurts GRPO; GRPO+teacher is the only cell that regresses in its second half — the maze safety reversal at LLM scale.

7.5 LLM scale: GSM8K 2×2, pre-registered (SmolLM2-360M, N=16, one A10G). Predictions committed before any cell finished. **P-G2, the riskiest prediction, confirmed:** GRPO+teacher is the only cell that regresses in its second half (val accuracy .096 → .093, pass@4 .193 → .181) while plain GRPO climbs monotonically (.078 → .105 → .120) — with the teacher verifiably steering (sampled-dead fraction driven to 0.48 vs. the ∼0.65 population rate). MaxRL+teacher trains stably to its best value (.066 → .102); the uniform-MaxRL cell finishes at .108, making P-G1 (a teacher gain for MaxRL) an honest *null at this budget* — predicted in advance by the posterior-starvation analysis below. A pass@k sweep on the final checkpoints sharpens the safety read: the teacher's deficit under GRPO *widens monotonically with* k (−.008 at k=1 to −.024 at k=16) — curriculum damage lives in coverage, as the maze predicted, though each individual delta is within single-seed noise. Beyond the predictions: the teacher's posterior *learns* real difficulty from ∼1 visit per prompt ($\rho = -0.17$ against independent 7B-model solve-rate annotations, $p \approx 10^{-17}$) but its allocation is *posterior-starved* — 3,200 draws over 7,473 prompts leave Thompson sampling near-uniform. And the GRPO degradation is not token-entropy collapse (GRPO+teacher retains *more* entropy): the damage is distributional, visible only in coverage. Single seed; small model; the pre-registered outcomes were orderings and signs, and both landed.

Takeaway. The safety half of the thesis transfers to LLM RLVR; the allocation half needs tiers or longer runs — which is how the next experiment is designed.

7.6 Countdown 2×2 (pre-registered): two honest nulls and a new phenomenon. The first exact-verifier hindsight experiment in RLVR ran in production (12 relabeled groups/step, 72–89% yield). Both positive predictions returned null, with an exact diagnosis: the uniform baseline reached 94/76/94% of the random-guesser bound implied by the pool's simplified construction — no headroom for allocation or creation to matter. What the run found instead is new: **recycling-induced sharpening** — hindsight arms improved mean@16 while *losing* pass@16 coverage (tier-1: .571 vs. .645 for the no-relabel baseline). Recycled off-target successes concentrate the policy on reachable outputs at the expense of exploration breadth: the creation channel has its own safety trade-off, mirroring §7.3's weight-induced collapse — data-induced rather than weight-induced, and to our knowledge unreported (no prior work measures pass@k under hindsight in RLVR). Our algebra predicts it: a relabel to a self-achieved value is a task at $p \approx 1$, where $u(p)$ says training signal is zero — the softmax pays for those updates out of the exploration tail. *Takeaway: recycling needs an* *admission rule, and the estimator already provides it — the same derived utility that schedules sampling should gate the relabel destination.*

8 Related work

Difficulty-adaptive RLVR sampling is bucket- or scalar-level and heuristic-scored: DUMP (UCB over buckets), AdaRFT (scalar controller over precomputed labels), SEC (category bandit), threshold curricula. LILO is the closest per-prompt method — rejection sampling by $\hat{p}(1-\hat{p})$, which Prop. 1's corollary identifies as the $N=2$ slice of our derived utility. DAPO's dynamic sampling and GRESO cull dead prompts — avoidance without allocation or creation. PKPO and Pass@k-training modify the objective toward pass@k but keep uniform sampling. Hindsight for LLMs (AgentHER, HSL) relabels agent trajectories with an LLM judge; exact-verifier relabeling inside the RL loop is, to our knowledge, unoccupied. No prior work (i) derives the sampling utility from the estimator's own expected signal, (ii) couples it with success-conditioned weights it provably matches, and (iii) adds a signal-creation channel with an exactness guarantee.

9 Limitations and honest negatives

Deep frontiers at fixed budget remain uncrossed on the maze, and a 4× duration run *refuted* our own duration hypothesis (the stall is a per-step-competence ceiling; depth needs capacity, not more schedule). γ-concentration did not transfer from chains to the maze — predicted in advance by the compounding ODE model. Adaptive truncation order underperformed fixed T. Feeding relabeled successes to the teacher's posterior inflates it — dropped. An early GSM8K claim of teacher steering was retracted when review found the sampler epoch-frozen; the fixed run confirmed the effect properly. Recycled-gradient exactness is a property of the environment's relabel map; noisier verifiers will land between our +0.22 (fixed pools) and +0.01 (one-shot) endpoints. LLM results are single-seed at 360M; the 2×2×2 and multi-seed replications are the active work.

Reproducibility

Every proposition has a Monte-Carlo verification script; every experiment writes JSON/JSONL artifacts committed to the repository; the documentation passed a two-round adversarial audit. The framework is a dependency-light package (`frontier_rl`, numpy-only core) with adapters for LLM pools (verl), gym control, IsaacLab parallel sim, and flow-policy VLAs.

References

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